AGRIMA SUDHIR

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Education

University Institute of Engineering and Technology, Panjab University, Chandigarh *2023 – 2027*
B.E. in Computer Science and Engineering *CGPA: 9.56/10*

Delhi Public School, Chandigarh *2021 – 2023*
Higher Secondary, Non-Medical Stream *97.4%*

Research Experience

Nottingham Trent University, UK *May 2024 – May 2025*
AI Research Intern – Medical Technologies Innovation Facility

- Conducted research on **deep learning for 3D medical image reconstruction**, designing the preprocessing pipeline and volumetric U-Net based neural architectures for cranial, mandibular, and maxillofacial defect reconstruction.
- Designed and evaluated experiments across model architectures including **quantitative benchmarking, ablation studies, and biomechanical validation**, in collaboration with an international multidisciplinary research team.
- Co-authored **2 peer-reviewed publications** arising from this research, contributing to the development & validation of AI-based approaches for cranial implant generation, reducing time taken from **240 to 25 secs** and attaining **0.99 DSC**.

Easiofy Solutions Pvt Ltd *June 2025 – Present*
Machine Learning Engineer – Medical AI

- Led development of a **deep learning system for 3D cranial defect reconstruction**, translated toward real-world clinical use through a partnership with a US-based healthcare technology company; **patent filed**.
- Built a **15,000+ sample 3D medical imaging dataset** combining synthetic and clinically-inspired defect patterns to support large-scale training and evaluation of volumetric reconstruction models.
- Architected a **multi-modal deep learning framework for radiotherapy dose prediction from 3D Head & Neck CT scans**, incorporating attention mechanisms and volumetric learning.

Design Innovation Centre (DIC), MDaRT Lab, Panjab University *Jan 2024 – Present*
Research Team Lead / Technical Consultant

- Led interdisciplinary research coordination with PGIMER and GMCH Hospitals, translating clinical requirements into technical research problems and development roadmaps for medical technology projects.
- Designed and led the annual summer research internship programme in the track **Full Stack Artificial Intelligence**, mentoring **80 students** and receiving the **Best Mentor Award (2026)**; represented the lab at the All India DIC Meet, IIT Delhi.

Patents & Publications

- **Patent Granted (India, No. 569335):** *Method for Generating Custom Cranial Implants Using Deep Learning* [Link]
- **Second Indian Patent Granted (India, No. 598648):** *A system for generating a cranial implant for a patient and its method thereof.*
- **Publication:** "MaxI-Net: A 3D AI Framework for CBCT-Based Maxillofacial Defect Reconstruction and Patient-Specific Implant Generation with Biomechanical Validation." *Bioengineering*, MDPI, 2026. DOI
- **Publication:** "SCAI-Net: AI-driven framework for optimized, fast, and resource-efficient skull implant generation." *Computers in Biology and Medicine*, Elsevier, Vol. 194, 2025. DOI
- **Publication:** "A Comprehensive AI Framework for Superior Diagnosis, Cranial Reconstruction, and Implant Generation." *Bioengineering*, MDPI, 2025. DOI
- **Open Source – LLM4S (Scala):** Contributing to LLM4S; enhanced CI/CD pipeline and backend utilities for LLM integrations.

Skills

Languages: Python, C, C++, SQL, Scala (learning).

ML / AI: PyTorch, 3D CNNs, U-Net, GANs, Attention Mechanisms, Transformer, Reinforcement Learning, Computer Vision, Medical Image Processing, Agentic AI Workflows, Ablation Studies.

Foundations: Linear Algebra, Probability & Statistics, Classical AI & Search (uninformed/informed search, CSPs, game trees), Deep Learning Theory, LLM Architectures & Evaluation.

Academic Validation: GATE CSE 2026: AIR 707 | GATE Data Science & AI 2026: AIR 848.

Data & Tools: Git, CUDA/GPU Training, pandas, NumPy, REST APIs, Linux.

Leadership: Research team lead, cross-institutional collaboration, technical mentoring, stakeholder management.